Head to Head Movement

V-to-T Raising

2026-11-09

2026HCS7009 · HSL7721

1 The problem

X-bar cannot draw this sentence

French	**English**
Je mange souvent des pommes.	I often eat apples.
I eat often of.the apples	I often eat apples.

In French, the adverb *souvent* sits between the verb and its object. In English, the adverb precedes the verb.

The object is the complement of V, a sister to V. X-bar theory predicts that nothing can intervene between a head and its complement. The French sentence is grammatical, so the theory undergenerates.

Undergeneration: the grammar fails to generate a grammatical sentence.

The adverb is an adjunct

An adverb is adjoined to V’ or VP. In the base tree it precedes the verb:

```
[V' souvent [V' mange des pommes]]
```

If the verb stays in V, the order is *souvent mange*, not *mange souvent*. Moving the verb to T past the adverb gives the French order, and the object stays in its complement position.

The adverb only makes the verb’s position visible. The analysis has to say where the tensed verb sits relative to T.

Transformations

Chomsky claimed that PSR cannot generate every sentence of a language. He proposed a second rule type: transformations. A transformation takes a structure built by X-bar rules and changes it in limited ways.

Lexicon + X-bar rules ("the base") → **D-structure** → **transformational rules** → **S-structure** → grammaticality

- D-structure is the output of the base, before any transformation. It is filtered by the Theta Criterion.
- S-structure is the output of the transformations, the string you say. It is filtered by the EPP.
- A trace marks the position the moved item left. It is not pronounced.

2 The rule

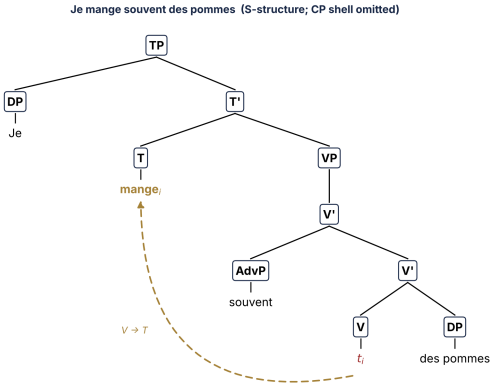
V → T

V → T movement: move the head V to the head T.

- Motivation: the verb needs its inflectional affix.
- Affix lowering is the mirror operation: T lowers onto V.
- The two operations are in complementary distribution. A language has one or the other.
- The rule is an instance of head-to-head movement.

The moved verb leaves a trace in V. The trace is not pronounced, but it marks the base position.

The French derivation



D-structure: *Je* [TP T [VP [V' *souvent* [V' *mange des pommes*]]]].

V ⟶ T moves *mange* to T.

S-structure: *Je mange souvent t des pommes.*

English lowers the affix

- The D-structure has the same shape.
- English does not raise the main verb.
- English lowers the tense affix from T onto V.
- S-structure: *He often eats apples.*
- The adverb stays before the verb, and the object stays next to V.

The two languages differ in the direction of the operation, not in the base structure.

The parameter

(9) Verb raising parameter: verbs raise to T, or T lowers to V.

- French sets the parameter to raising.
- English sets it to lowering.
- This is Carnie's first statement of the parameter.
- The parameter is supposed to be a binary choice.

3 Evidence beyond French

English auxiliaries

	English	**French**
main verb + adverb	I often eat apples	Je mange souvent des pommes
auxiliary + adverb	I have often eaten apples	J'ai souvent mangé des pommes
main verb + negation	I do not eat apples	Je ne mange pas des pommes
auxiliary + negation	I have not eaten apples	Je n'ai pas mangé des pommes

Auxiliaries are T. They precede the adverb and negation in both languages. French main verbs do too. English main verbs follow the adverb and negation.

English auxiliaries behave like French main verbs: they raise.

The parameter splits

The first statement cannot cover English. English auxiliaries raise, but English main verbs lower. Carnie revises the parameter:

(46) Revised verb movement parameter
Option 1: all tensed verbs raise to T.
Option 2: tensed auxiliaries raise to T and T lowers to tensed main verbs. English chooses option 2.

The new rule is no longer a single binary switch. Option 2 bundles two settings: auxiliaries raise, main verbs lower.

Vata: V → T in an SOV language

Data from Koopman (1984), as given in Carnie, Chapter 9.

Vata	**Gloss**	**Translation**
A la saka li.	we have rice eaten	We have eaten rice.
A li saka.	we eat rice	We eat rice.

- With an auxiliary (*la*), the verb stays at the right edge.
- Without an auxiliary, the verb raises around the object into T.
- The base is head-final, but the verb still moves.
- Head movement is independent of the OV/VO order parameter.

4 Problems

Problem 1: the parameter is disjunctive

- English auxiliaries raise whereas English main verbs lower.
- The revised parameter is a disjunction: option 1 or option 2, and option 2 contains two sub-options.
- A disjunctive parameter weakens the claim that parameters are simple.

Problem 2: affix lowering is stipulated

- Carnie: the negative morpheme *not* blocks affix lowering. “The reasons for this are obscure.” He states it as a stipulation.
- The theory says English T lowers onto V, but it does not explain why *not* stops that operation.
- Do-support is the escape hatch, but it is an insertion rule of last resort. It does not explain the blocking either.
- The account is general at the level of the rule and local at the level of the exception.

5 Summary

Summary

- X-bar undergenerates. Transformations add restricted changes.
- V $\longrightarrow$ T raises the verb to T to support inflection. Affix lowering is the mirror.
- The rule handles French and Vata. English needs the revised parameter.
- The cost: a disjunctive parameter, a stipulation about negation, and no diagnostic in SOV languages such as Malayalam.

References

- Carnie, A. (2013). *Syntax: A Generative Introduction*. John Wiley and Sons. Chapter 9.
- Chomsky, N. (1957). *Syntactic Structures*. Mouton.
- Koopman, H. (1984). *The Syntax of Verbs: From Verb Movement Rules in the Kru Languages to Universal Grammar*. Foris.
- Pollock, J.-Y. (1989). Verb movement, Universal Grammar, and the structure of IP. *Linguistic Inquiry*, 20, 365-424.
- Lightfoot, D. & Hornstein, N. (eds.) (1994). *Verb Movement*. Cambridge University Press.